

Nallabati Venkata Durga Sai Sarat Chandra

Target Role: Data Analyst Intern | Swiggy

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SUMMARY

Visionary Robotics & AI undergraduate specializing in the intersection of edge mechatronics, Scientific Machine Learning, and human-centric design. Combines hands-on development of autonomous hardware and advanced physics-informed models with intuitive software architecture. Poised to deliver high-impact, user-friendly edge intelligence systems that seamlessly bridge complex physical dynamics with operational clarity.

EDUCATION

Bachelor of Technology (B.Tech), Amrita Vishwa Vidyapeetham CGPA: 9.05 / 10.00	2024 – Present Amritapuri
Higher Secondary Education, Sri Chaitanya Vidhyanikethan CBSE (MPCEA) Percentage: 88.2%	2024 Vishakapatnam
Secondary School Education, Sri Chaitanya School SSC AP Marks: 570 / 600	2022 Rajam

SKILLS

Languages & Runtimes: Python (3.12), SQL (PostgreSQL), C++ (C++17/20), Bash, MATLAB, POSIX Sockets
Data Science & Machine Learning: Pandas, NumPy, Scikit-Learn, Optuna (Bayesian Optimization), PyTorch, LightGBM, XGBoost, Conformal Prediction
Data Engineering & Pipelines: Time-Series DSP Feature Extraction, ETL Ingestion, REST APIs, JSON/Parquet, SVD/POD Modal Decomposition
Tooling, Simulation & Systems: Jupyter, Linux (Ubuntu Server), Docker, Git/GitHub, LaTeX, ANSYS Fluent (CFD), Statistical Coverage Testing

PROJECTS

Domain-Aware Industrial Failure Prediction & Diagnostics Framework Pandas, NumPy, Optuna Bayesian Ensemble, Conformal Uncertainty, XGBoost • 85.2% recall at 91.4% precision (0.94 AUROC) across 120k telemetry time-series samples.	PIML Researcher / Data Science Lead
Blood Neural Network: PINNs for Inverse Hemodynamic Rheology Discovery PINNs, Navier-Stokes, Carreau-Yasuda, ANSYS Fluent CFD, NTK Trace Balancing • <2.4% relative L2 error vs high-resolution ANSYS Fluent pulsatile CFD benchmarks.	SciML Researcher
Multi-Parameter Health Vitals Monitor & Real-Time IoT Data Streamer MAX30102 PPG, ESP8266/ESP32, Fast-Mode I2C, 2nd-order IIR Bandpass, FreeRTOS • <200ms real-time SpO2/HRV cloud telemetry; 9-clock I2C bus lockup recovery.	Biomedical Systems Developer

PUBLICATIONS

Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity, IEEE ICRM 2025 • Architecting Level 4 autonomous navigation logic within a 2.4 GHz CPU hardware bottleneck, utilizing Probabilistic Hough Transforms for real-time computer vision processing, and engineering a handshake-free UDP telemetry protocol to eliminate multi-agent V2V latency in edge environments.	11/2025
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ACHIEVEMENTS

- Selected among top international teams to complete all three technical milestones of the IEEE IES Generative AI Challenge.
- Successfully cleared the 2 Levels of NAEST (India's premier physics experimental competition organized by IAPT and Dr. H.C. Verma).
- Selected for the ISRO-IIRS YUVIKA through a national-level merit screening and science aptitude evaluation.
- Qualified for MTSO (Maths Olympiad) Level 2 (Stage II) by ranking in the top tier of candidates nationwide.